

Topic colour: first scope, in this order, among the current courses teaching it (by teachers' and programmes' department)
service: some DISI teacher, no DISI degree programme
external: no DISI teacher, no DISI degree programme
borrow: no DISI teacher, some DISI degree programme
weak_internal: some (not all) DISI teachers and DISI degree programmes
internal: only DISI teachers and DISI degree programmes

Computing and Digital Information Foundations
6 ECTS
computer_system_fundamentals
digital_information_representation
digital_information_and_software_levels
binary_number_representation
computing_history
programming_language_evolution

Programming Concepts
6 ECTS
programming_concepts_introduction
imperative_programming_constructs
functions_and_recursion
program_input_output_and_files
string_processing
development_environment

Programming Data Structures
6 ECTS
fundamental_data_structures
abstract_data_types
memory_organization_and_pointers
program_organization_and_libraries
debugging
exception_handling

Object-Oriented Programming
6 ECTS
object_oriented_programming
object_oriented_design_principles
java_data_structure_programming
java_application_features
software_design_patterns
component_oriented_programming

C and C++ Programming
6 ECTS
c_programming
cpp_programming_features
memory_organization_and_pointers
program_input_output_and_files
program_organization_and_libraries
string_processing

System Programming
3 ECTS
system_programming

Algorithms, Data Structures and Complexity
12 ECTS
algorithm_design_analysis
algorithm_cost_analysis
computational_complexity_theory
computability_models_and_limits
sorting_selection_priority_structures
hashing_disjoint_set_structures
graph_algorithms
recursive_divide_conquer_algorithms
dynamic_programming
greedy_algorithms
exact_search_techniques
heuristic_approximation_search

Bioinformatics Search and String Algorithms
3 ECTS
string_processing_structures_algorithms
biological_search_tasks

Python Programming
6 ECTS
python_programming_core
scripting_languages
python_regular_expressions
notebook_computing
python_project_packaging
python_web_page_processing

Scientific and Numerical Programming
9 ECTS
scientific_computing
numerical_analysis_and_errors
numerical_modeling
matlab_octave_programming
python_data_science_tools
open_source_and_linux_tools
python_turtle_graphics